1.

Let $\mathcal{T} = \langle S, L, c, T, s_0, S_* \rangle$ be the transition system depicted by the graph G.

(a)

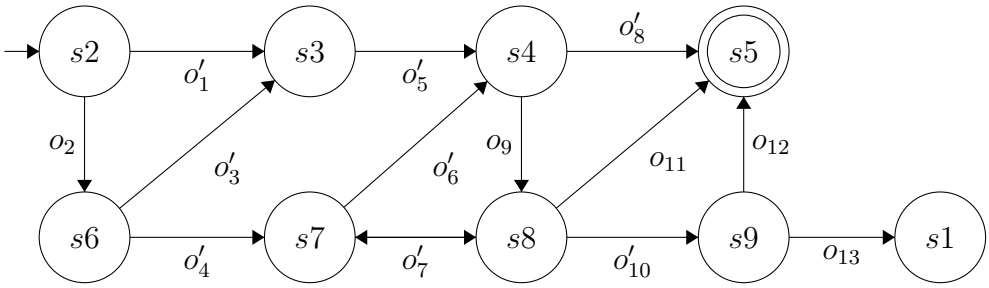


Figure 1: The graph G_1, depicting the transition system T^1.

The graph G_1, shown in Fig. 1, depicts a transition system $\mathcal{T}^1 = \langle S^1, L, c, T^1, s_0^1, S_*^1 \rangle$ obtained by applying a function $\varphi^1 : S \to S^1$ to the set of states of the original transition system (T^1, s_0^1 and S_*^1 are modified accordingly).

φ^1 is defined as $\varphi^1(s_i) = s_{(i \bmod 9)+1} \quad \forall s_i \in S$.

Since φ^1 is a bijective function, and since we did not changed the original operators (which can be seen as applying the identity function to L), this new transition system $\mathcal{T}^1$ is isomorphic to $\mathcal{T}$. Hence, G_1 is isomorphic to G.

G_1 is, however, not the same graph as G, since their adjacency matrices are distinct to each other.

(b)

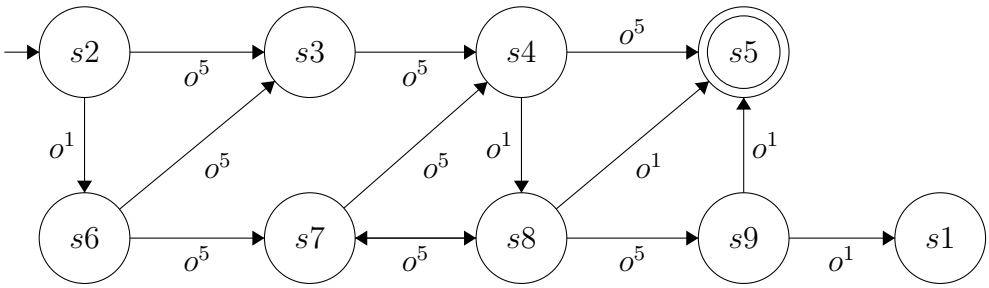


Figure 2: The graph G_2, depicting the transition system T^2.

The graph G_2, shown in Fig. 2, depicts a transition system $\mathcal{T}^2 = \langle S^2, L^2, c^2, T^2, s_0^2, S_*^2 \rangle$ obtained by applying a function $\varphi^2 : S \to S^2$ to the set of states of the original transition system, and a function $\lambda^2 : L \to L^2$ to the set of operators of the original transition system (T^2, s_0^2 and S_*^2 are modified accordingly), as well as defining a new cost function $c^2 : L^2 \to \mathbb{R}_0^+$ for the new set of operators L^2.

The function φ^2 is defined exactly the same way as φ^1 (from item (a)).
The function λ^2 is defined as

$$\lambda^2(o) = \begin{cases} o^5, & \text{if c(o)} = 5, \\ o^1, & \text{otherwise.} \end{cases} \quad \forall o \in L \tag{1}$$

And the cost function c^2 is defined as $c^2(\lambda^2(o)) = c(o) \quad \forall o \in L$.
$\mathcal{T}^2$ is graph-equivalent to $\mathcal{T}$ because the costs of the transitions are preserved and φ^2 is the exact same function as φ^1 from item (a) (which resulted in an isomorphic transition system in that case). However, since λ^2 is not bijective, the graph G_2 is not isomorphic to G.

(c)

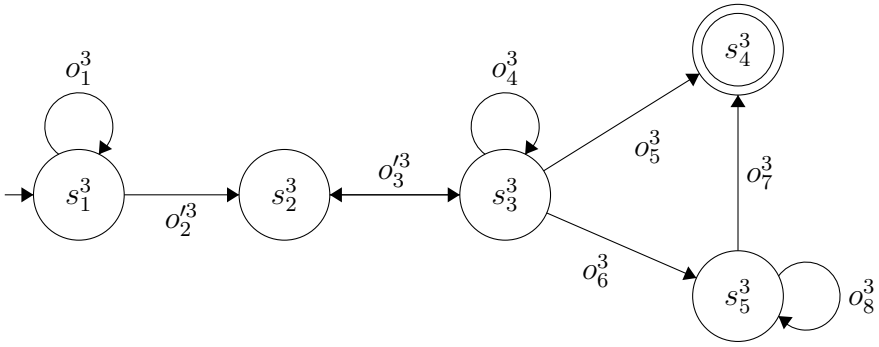


Figure 3: The graph G_3, depicting the transition system T^3.

The graph G_3, shown in Fig. 3, depicts a transition system $\mathcal{T}^3 = \langle S^3, L^3, c^3, T^3, s_0^3, S_*^3 \rangle$ obtained by applying a function $\varphi^3 : S \to S^3$ to the set of states of the original transition system, and a function $\lambda^3 : L \to L^3$ to the set of operators of the original transition system (T^3, s_0^3 and S_*^3 are modified accordingly), as well as defining a new cost function $c^3 : L^3 \to \mathbb{R}_0^+$ for the new set of operators L^2.
φ^3 is such that $\varphi^3(s_1) = \varphi^3(s_5) = s_1^3,\ \varphi^3(s_2) = \varphi^3(s_6) = s_2^3,\ \varphi^3(s_3) = \varphi^3(s_7) = s_3^3,\ \varphi^3(s_4) = s_4^3,\ \varphi^3(s_8) = \varphi^3(s_9) = s_5^3$.
λ^3 maps each operator in the original transition system depicted by an edge that goes from state s to state s' ($s, s' \in S$) to a new operator depicted by an edge going from state $s^3 = \varphi^3(s)$ to state $s'^3 = \varphi^3(s')$. The cost of this new operator is the minimum cost between all operators that goes from a state in the set of all states that are mapped to s^3 by φ^3 to a state in the set of all states that are mapped to s'^3 by φ^3.
Since, for all pair of states s and s' connected by an edge representing an operator o in the original transition system, we have an equivalent edge $\lambda^3(o)$ connecting $\varphi^3(s)$ and $\varphi^3(s')$, and the initial and goal states of the original transition system are mapped to the initial and goal states of this new transition system by the function

φ^3, respectively, the graph G_3 is strictly homomorphic to G. However, G_3 is not graph-equivalent to G, since $\varphi^3(s')$ is not a bijective function.

(d)

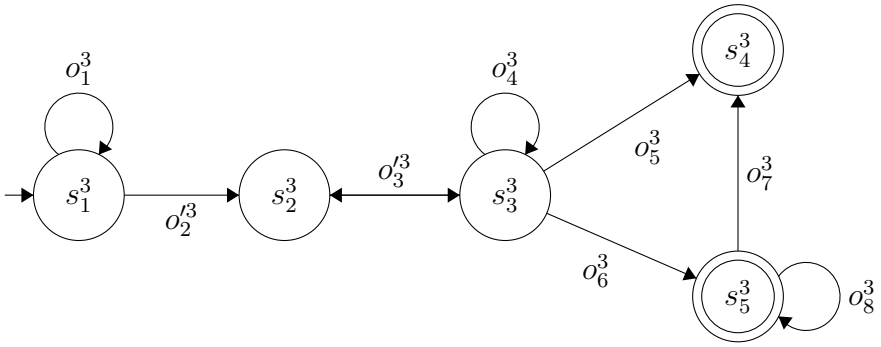


Figure 4: The graph G_4, depicting the transition system T^4.

The graph shown in Fig. 4 is built similarly as the one from item (c), but with one additional goal state. The inclusion of this additional goal state makes the transition system T^4 depicted by G_4 makes it a non-strict homomorphism of the original transition system.

(e)

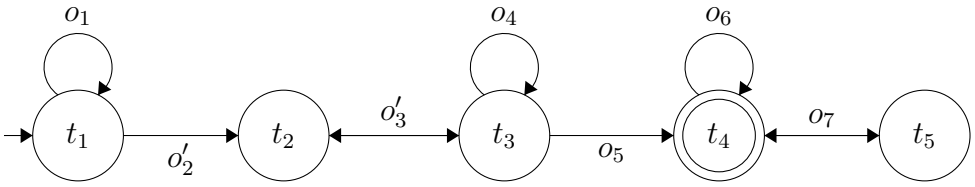


Figure 5: The graph G_5, depicting the transition system T^5.

The graph G_5, shown in Fig. 5, depicts a transition system $\mathcal{T}^5 = \langle S^5, L^5, c^5, T^5, s_0^5, S_*^5 \rangle$ obtained by applying a function $\alpha : S \to S^5$ to the set of states of the original transition system, and a function $\lambda^5 : L \to L^5$ to the set of operators of the original transition system (T^5, s_0^5 and S_*^5 are modified accordingly), as well as defining a new cost function $c^5 : L^5 \to \mathbb{R}_0^+$ for the new set of operators L^5.

This graph is a strict homomorphism of the graph depicting the original transition system because, for all pair of states s and s' connected by an edge representing an operator o in the original transition system, we have an equivalent edge $\lambda^5(o)$ connecting $\alpha(s)$ and $\alpha(s')$. Additionally, the initial and goal states of the original transition system are, respectively, mapped to the initial and goal states of this new transition system by the function α. Since G_5 is strictly homomorphic to G, α defines

an abstraction of the original transition system $\mathcal{T}$, hence, the resulting transition system $\mathcal{T}^5$ (or, using the standard notation for abstract transition systems, $\mathcal{T}^\alpha$) is the abstract transition system induced by α over $\mathcal{T}$.

(f)

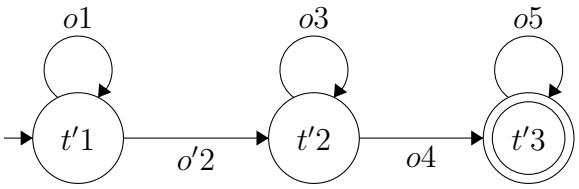


Figure 6: The graph G_6, depicting the transition system T^6.

The graph G_6 is the induced transition system of an abstraction $\beta = \phi \circ \alpha$, where ϕ is an abstraction that maps the sets of states $\{t_1, t_2\}, \{t_3\}, \{t_4, t_5\}$ of G_5 into, respectively, the states t'_1, t'_2 and t'_3 of G_6. Since G_6 is strictly homomorphic to G_5, ϕ is an abstraction that maps the transition system depicted by G_5 into the (abstracted) transition system depicted by G_6. Hence, since abstractions are composable, $\beta = \phi \circ \alpha$ is an abstraction, and the graph G_6 is the graph depicting an abstracted transition system induced by the β applied over the original transition system $\mathcal{T}$. β is a coarsening of α since it is defined by a composition of another abstraction ϕ with α.

(g)

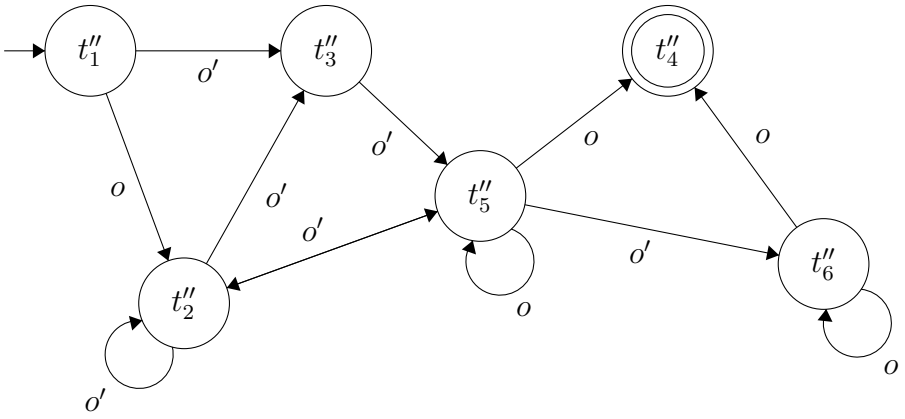


Figure 7: The graph G_7, depicting the transition system T^7.

The graph G_7 is the induced transition system of an abstraction ϕ' applied over the original transition system $\mathcal{T}$.

ϕ' maps the sets of states $\{s_1\}, \{s_5, s_6\}, \{s_2\}, \{s_4\}, \{s_3, s_7\}, \{s_8, s_9\}$ of the graph G to, respectively, the states $t''_1, t''_2, t''_3, t''_4, t''_5$, and t''_6 of the graph G_7. This map preserved the initial state (s_1 is mapped to t''_1), the goal state (s_4 is mapped to t''_4), and all

connections (for each edge between s, s' on G, there is an edge between $\phi'(s)$ and $\phi'(s')$ in G_7 with cost equal to the minimal cost between any pair of nodes u, v in G mapped to $\phi'(s), \phi'(s')$, hence, it is an abstraction.

The abstraction ϕ' can be composed with another abstraction ϕ'' to form the abstraction β. ϕ'' maps $\{t''_1, t''_2, t''_3\}$ into t'_1, t''_5 into t'_2, and $\{t''_4, t''_6\}$ into t'_3. Therefore, ϕ' is a refinement of β.